

Brainstorming & Idea Prioritization

Date	28 September 2026
Team ID	6ab9f220c23107aaaa73f984
Project Name	ComicCraft AI - AI comic story creator
Maximum Marks	3 Marks

Step 1: Brainstorm and idea listing

Feature opportunities were derived from the supplied ComicCraft source code and project documentation, then grouped by the user need they address.

S.No	Team member	Idea / suggestion	Category	Group no.
1	Arun N	Genre-specific starter prompts and examples	Onboarding	1
2	Pushkaran P	Character reference support for visual continuity	Generation quality	2
3	Pravin R	Progress updates while the five panels generate	Transparency	1
4	Bhuvanesh P	Targeted redraw for a single weak panel	Creator control	2
5	Jayaraj	Editable dialogue and captions before export	Editing	3
6	All members	Export choices and a reusable comic history	Sharing	3

Step 2: Idea prioritization

Grouped ideas are rated for implementation feasibility and user importance. The priority order identifies a practical product-improvement sequence.

Group no.	Final idea	Feasibility	Importance	Priority	Selected
1	Guided creation: starter prompts + generation progress	High	High	1	Yes
2	Consistent artwork: character references + panel redraw	Medium	High	2	Yes
3	Creator finishing tools: editable text + export history	High	Medium	3	Yes

Define Problem Statements

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Define Problem Statements (Customer Problem Statement Template)

A customer problem statement keeps the team focused on what matters to the people ComicCraft is built for. Each row reads: *I am ... I'm trying to ... But ... Because ... Which makes me feel ...*

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A hobbyist storyteller with a story idea	Turn my idea into an illustrated comic	I can't draw and don't know comic layout	Hiring an artist is slow and expensive, and drawing tools have a steep learning curve	Frustrated, but excited if it could be done in minutes
PS-2	A teacher preparing lesson material	Make a short illustrated story to explain a topic	I have very little time to prepare visuals	Finding or drawing matching images for each step takes hours	Overwhelmed, hopeful for a quick ready-made option
PS-3	A student or content creator	Create a comic with the same character in every panel	AI images change the character's look from panel to panel	Most tools generate each image independently, and text inside images comes out garbled	Disappointed with the result, wanting control
PS-4	A user who likes the story but not one panel	Fix a single bad panel	I have to regenerate the whole comic	Many tools have no per-panel redraw	Annoyed at wasting time, relieved if I can redo one panel

Empathy Map

Date	28 September 2026
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Maximum Marks	4 Marks

Empathy Map

Empathy map for the primary user of ComicCraft, showing what they Say, Think, Do and Feel in relation to the problem:
turning a story idea into an illustrated comic without drawing skills.

SAYS • "I have a great story but I can't draw." • "Can it keep my character looking the same?" • "Can I just redo this one panel?"	THINKS • "Will the AI actually understand my idea?" • "Hiring an artist is too costly for a hobby project." • "Will the text in the pictures be readable?"
DOES • Searches for comic makers and AI art tools • Types a prompt, character, setting, tone and art style • Regenerates panels and downloads the PDF to share	FEELS • Excited about seeing the story come alive • Frustrated when characters change between images • Proud and satisfied when the final comic looks good

RIYA
*Aspiring
 Storyteller
 (Student /
 Hobbyist Creator)*

Persona name is illustrative.

Customer Journey Map

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Customer Journey Map

Journey of a user from discovering ComicCraft to sharing a finished comic.

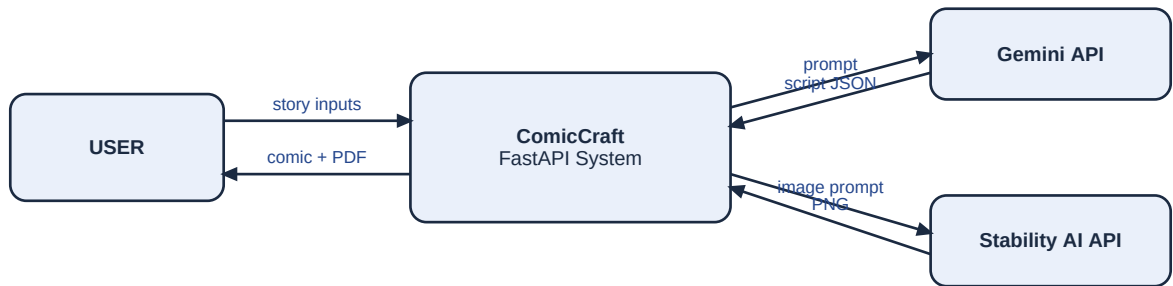
Stage	User action	Touchpoint	Emotion	Pain point	Opportunity
1. Discover	Hears about an AI comic creator	Web link / demo	Curious	Unsure it works for their own story	Clear tagline and demo
2. Describe	Types story, character, setting; picks tone and art style	index.html form	Creative, engaged	Blank-page problem	Placeholder examples, dropdown choices
3. Generate script	Clicks Generate Comic	POST /generate-comic (Gemini Flash)	Anticipation	Waiting with no feedback	Progress bar and status text
4. Watch panels appear	Sees 5 panels drawn one by one	POST /generate-image (SDXL)	Delighted or impatient	Slow image calls, occasional failures	Placeholder panel, per-panel retry
5. Review and refine	Reads headings, narration, dialogue; redraws weak panels	Comic card layout, Redraw button	In control	A panel looks off	Redraw only that panel
6. Export	Downloads PDF storyboard	POST /download-pdf, export page	Proud, satisfied	PDF layout mismatch	Consistent ReportLab layout
7. Share	Shares PDF with friends, class or social media	PDF file	Happy	Does not know how to start the next story	Easy 'create another comic' action

Data Flow Diagram

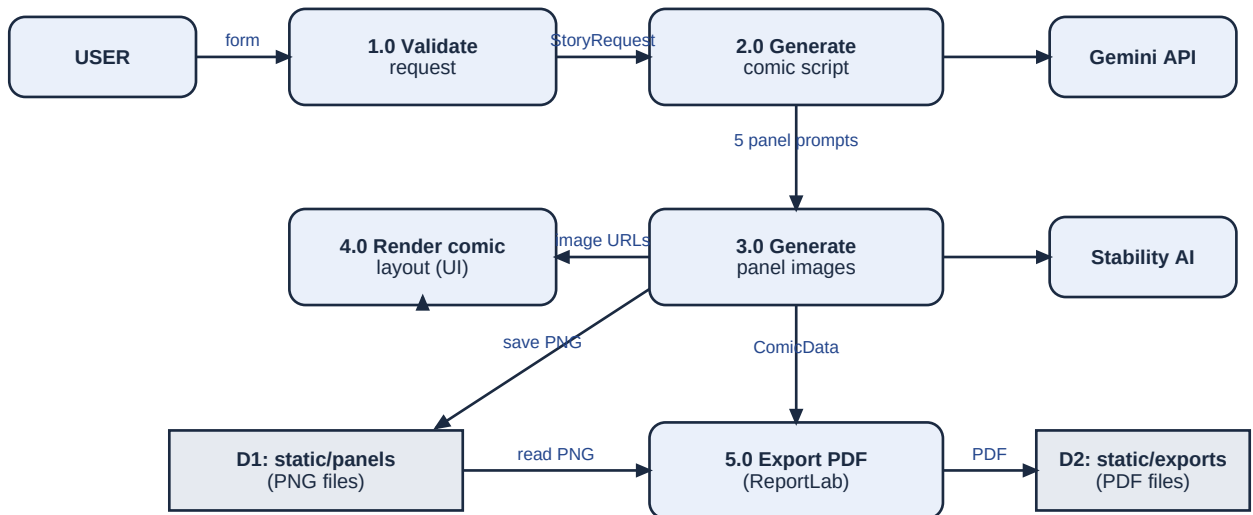
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Data Flow Diagram

Level 0 (context diagram)



Level 1 (processes and data stores)



Data stores and flows

- **Input:** prompt, character, setting, tone, art style (StoryRequest model).
- **Process 2.0:** Gemini returns a title and exactly 5 panels (heading, scene, action, expression, dialogue, caption, image prompt) as JSON.
- **Process 3.0:** each image prompt is sent to Stable Diffusion XL; the PNG is stored in static/panels and its URL returned.
- **Process 5.0:** ComicData is laid out with ReportLab and saved in static/exports; the PDF URL is returned to the user.

Solution Requirements

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Functional Requirements

ID	Requirement	Priority
FR-1	User can enter a story idea, main character and setting	High
FR-2	User can pick a story tone (Funny, Adventurous, Dark, Heartwarming, Sci-Fi, Mystery, Epic)	High
FR-3	User can pick an art style (Classic Comic Book, Manga, Watercolor, Pixel Art, Noir, Cartoon, Retro Pop Art)	High
FR-4	System generates a comic title and exactly 5 connected panels via Gemini	High
FR-5	Each panel has heading, narration, dialogue, caption and an image prompt	High
FR-6	System generates one image per panel with Stable Diffusion XL	High
FR-7	Dialogue is rendered as a speech bubble over the image by the web page	Medium
FR-8	User can redraw an individual panel	Medium
FR-9	User can export the finished comic as a PDF	High
FR-10	A placeholder panel is used when image generation fails	Medium

Non-Functional Requirements

ID	Category	Requirement
NFR-1	Security	API keys are stored in backend .env and never sent to the browser; .env is excluded from Git
NFR-2	Usability	Single-page form with progress feedback and clear error messages
NFR-3	Responsiveness	Layout adapts to desktop and smaller screens
NFR-4	Reliability	One failed panel must not stop the rest of the comic
NFR-5	Maintainability	Separate modules for routes, models, script generation, images and export
NFR-6	Portability	Runs locally on Python 3.10+ with no Node.js build step
NFR-7	Performance	Script returned in seconds; image generation is the slowest step and is done per panel

Technology Stack

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Technology Stack

Layer	Technology	Purpose
Story generation	Google Gemini Flash (google-genai SDK)	Title, panels, dialogue, captions, image prompts as structured JSON
Image generation	Stable Diffusion XL 1.0 via Stability AI API	Comic panel artwork (stable-diffusion-xl-1024-v1-0)
Backend	Python 3.10+, FastAPI, Uvicorn	REST endpoints and page serving
Validation	Pydantic	Request and comic data models
PDF export	ReportLab	Storyboard PDF
Image handling	Pillow	Image validation and placeholder panels
HTTP client	requests	Calls to the Stability AI API
Frontend	HTML5, CSS, vanilla JavaScript	Form, progress, comic layout, redraw
Templating	Jinja2	index.html, comic_preview.html, export_success.html
Config	python-dotenv	Loads keys and settings from .env

Architecture note

The app uses a two-stage AI pipeline: Gemini writes the script, then Stable Diffusion XL draws each panel. Text is kept out of the images on purpose; the web page draws dialogue and captions so they stay readable.

Proposed Solution

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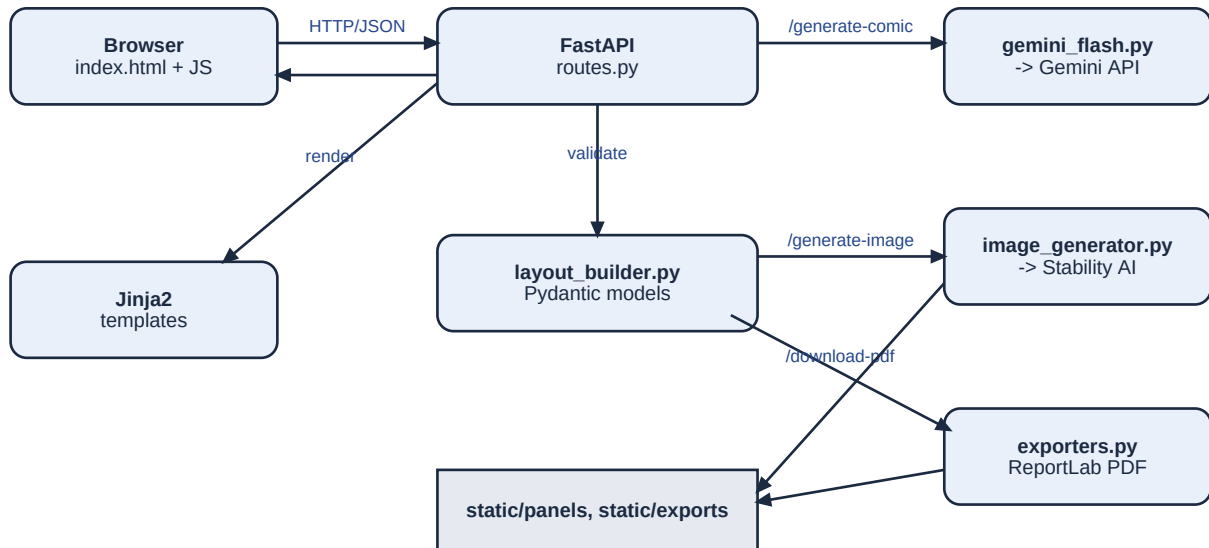
Proposed Solution

Item	Description
Problem	People with story ideas but no drawing skills cannot easily produce an illustrated comic.
Solution	A web app that turns a story prompt, character, setting, tone and art style into a 5-panel comic with artwork, narration, dialogue, captions and a PDF export.
Novelty	Two-stage AI pipeline; text kept out of images and drawn by the UI; per-panel redraw.
Value	A finished comic in minutes with no design skill; works for any story, not a fixed example.
Scalability	Stateless FastAPI endpoints; image and script calls can be queued or cached later.

Solution Architecture

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Solution Architecture



- The browser calls three JSON endpoints and renders panels, dialogue bubbles and the redraw control.
- API keys live only in the backend .env; the browser never sees them.
- Generated PNGs and PDFs are stored under static/ and served back by URL.

Project Planning

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Project Planning (Epics and Sprints)

Sprint	Epic	Tasks	Outcome
Sprint 1	Pre-requisites; Workflow; Model selection and architecture	Install Python and dependencies, get Gemini and Stability API keys, define the two-stage workflow, choose models	Environment ready, architecture agreed
Sprint 2	Core functionalities (FastAPI backend); routes.py	Pydantic models, Gemini script generation, SDXL image generation, ReportLab export, API routes	Working backend endpoints
Sprint 3	Frontend (Jinja2 templates)	index.html form, progress bar, comic card layout, dialogue bubbles, redraw, PDF button	Usable interface
Sprint 4	Testing and local deployment; Conclusion	Functional and performance testing, fix bugs, run locally with Uvicorn, final documentation and demo	Tested, documented project

Add your own start/end dates and story points per sprint if your template requires them.

Code Layout, Readability and Reusability

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Code Layout, Readability and Reusability

File	Lines	Responsibility	Reuse / readability
app/main.py	49	FastAPI entry point, CORS, static mount, router include	Single place for app setup
app/routes.py	114	Page routes and 3 JSON endpoints	Thin handlers; logic lives in modules
app/layout_builder.py	106	Pydantic models: StoryRequest, PanelData, ComicData, ImageGenRequest, PdfExportRequest	Shared data contract for backend and frontend
app/gemini_flash.py	335	Gemini client and structured script generation	Swap model through GEMINI_MODEL
app/gemini_pro.py	9	Optional narrative refinement hook (currently pass-through)	Extension point
app/image_generator.py	268	SDXL calls, style presets, placeholder fallback	Style map is a plain dict; model set via .env
app/exporters.py	377	ReportLab PDF layout	Independent of the web layer
templates/index.html	762	Main UI, CSS and vanilla JS	Functions: generate, drawPanel, redrawPanel, renderComicLayout, PDF download
templates/comic_preview.html, export_success.html	46 / 62	Standalone preview and export success pages	Small, single-purpose

Good practices used

- One module per concern (routes, models, script, images, export).
- Typed Pydantic models and docstrings on modules and functions.
- Configuration through environment variables; .env.example provided, .env ignored by Git.
- Errors are logged and returned as HTTP 500 with a message instead of crashing the server.
- Image failures fall back to a placeholder so the comic still renders.

Coding and Solution

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Coding and Solution

Selected excerpts from the actual source code. The full code is in the 'project source-code' folder.

Data models (app/layout_builder.py)

```
class StoryRequest(BaseModel):
    """Input sent by the user to create a new comic."""

    prompt: str = ""
    character: str = ""
    setting: str = ""
    tone: str = "Funny"
    art_style: str = "Classic Comic Book"

class PanelData(BaseModel):
    """Data for one comic panel."""

    panel_number: int = 1

    # Short title displayed directly below the image.
    panel_heading: str = ""

    # Main story/narration shown below the heading.
    scene_description: str = ""

    # Additional visual information used when building image prompts.
    character_action: str = ""
    character_expression: str = ""

    # Dialogue is rendered as a speech bubble by the frontend.
    dialogue: str = ""

    # Short dramatic caption displayed at the bottom.
    caption: str = ""

    # Prompt sent to Stable Diffusion XL.
    image_prompt: str = ""

    # Relative URL of the generated panel image.
    image_url: Optional[str] = None

    # True when image generation failed.
    failed: bool = False

class ComicData(BaseModel):
    """Complete generated comic."""

    title: str = ""
    panels: List[PanelData] = Field(default_factory=list)
    tone: str = "Funny"
    art_style: str = "Classic Comic Book"

class ImageGenRequest(BaseModel):
    """Request used by the image-generation endpoint."""

    image_prompt: str = ""
    art_style: str = "Classic Comic Book"
    panel_number: int = 1
    comic_id: Optional[str] = "default"
```

```
class PdfExportRequest(BaseModel):
    """Request used by the PDF export endpoint."""

    title: str = ""
    panels: List[PanelData] = Field(default_factory=list)
```

API endpoint: generate comic (app/routes.py)

```
try:
    comic = generate_comic_script(request)
    return comic.model_dump()
except Exception as e:
    logger.error(f"Error generating comic: {e}", exc_info=True)
    raise HTTPException(status_code=500, detail=str(e))

@router.post("/generate-image")
async def generate_image_endpoint(req: ImageGenRequest):
    """
    Generates one panel image using Hugging Face Inference Providers.
    Returns image URL.
    """
```

API endpoint: generate panel image (app/routes.py)

```
    image_url = generate_panel_image(
        image_prompt=req.image_prompt,
        art_style=req.art_style,
        panel_number=req.panel_number
    )
    return {"image_url": image_url}
except Exception as e:
    logger.error(f"Error generating panel image: {e}", exc_info=True)
    raise HTTPException(status_code=500, detail=str(e))

@router.post("/download-pdf")
async def download_pdf_endpoint(req: PdfExportRequest):
    """
    Builds the ReportLab PDF for the generated comic.
    Returns the downloadable URL.
    """
```

Style presets (app/image_generator.py)

```
STYLE_PRESETS = {
    "Classic Comic Book": "comic-book",
    "Manga": "anime",
    "Watercolor": "digital-art",
    "Pixel Art": "pixel-art",
    "Noir": "cinematic",
    "Cartoon": "digital-art",
    "Retro Pop Art": "comic-book",
}
```

No. of Functional Features Included in the Solution

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Number of Functional Features Included in the Solution

Total functional features: **12**

#	Feature	Description
1	Any story prompt	Natural-language story idea, not limited to a preset example
2	Character, setting, tone, art style inputs	7 tones and 7 art styles to choose from
3	5-panel story	Connected beginning-to-ending comic from Gemini
4	Panel headings, narration, captions	Text blocks shown under each image
5	Dialogue bubbles	Drawn over the image by the web page
6	AI artwork	Stable Diffusion XL image per panel
7	Panel redraw	Regenerate one panel
8	PDF export	Downloadable storyboard
9	Progress and error display	Progress bar and error box in the UI
10	Placeholder fallback	Panel placeholder if image generation fails
11	Responsive UI	Comic-card layout adapts to screen size
12	API key protection	Keys kept in backend .env

Performance Testing

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Performance Testing

Test plan with metrics to measure on the running application. Results are left blank on purpose: they must be recorded from your own runs with your API keys, so no numbers are made up here.

ID	Scenario	Metric	Target	Result
PT-1	POST /generate-comic with a typical prompt	Response time	Under 15 s	
PT-2	POST /generate-image for one panel	Response time	Under 30 s	
PT-3	Full flow: script plus 5 panels	Total time	Under 2.5 min	
PT-4	POST /download-pdf with 5 panels	Response time	Under 5 s	
PT-5	Redraw a single panel	Response time	Under 30 s	
PT-6	Page load of /	Load time	Under 2 s	
PT-7	3 users generating at once	Errors / slowdown	No failed requests	
PT-8	Image API failure (wrong key)	Behaviour	Placeholder shown, no crash	
PT-9	Missing GEMINI_API_KEY	Behaviour	Clear error message	

How to measure

- Browser DevTools, Network tab: note the time of each request.
- Uvicorn console logs for server-side errors.
- Repeat each test 3 times and record the average.

Project Executable Files

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Project Executable Files

Files needed to run ComicCraft locally.

File / folder	Purpose
app/main.py	Starts the FastAPI server
app/*.py	Routes, models, Gemini, image and PDF modules
templates/	index.html, comic_preview.html, export_success.html
static/	panels/, exports/, fonts/ (created at runtime)
requirements.txt	Python dependencies
.env.example	Template for API keys (copy to .env)

Run commands

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1      (Windows)
pip install -r requirements.txt
copy .env.example .env           (then add your keys)
python -m uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

Open <http://127.0.0.1:8000> in a browser.

Sample Project Documentation

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Sample Project Documentation

1. Overview

ComicCraft AI turns a story idea into a 5-panel illustrated comic with narration, dialogue, captions and a PDF export, using Gemini for the script and Stable Diffusion XL for the artwork.

2. Setup

Python 3.10+, a Google Gemini API key and a Stability AI API key. Install with `pip install -r requirements.txt`.

3. Configuration (.env)

```
GEMINI_API_KEY=your_gemini_api_key  
GEMINI_MODEL=gemini-3.8-flash
```

```
STABILITY_API_KEY=your_stability_api_key  
STABILITY_MODEL=stable-diffusion-xl-1024-v1-0
```

```
HOST=127.0.0.1  
PORT=8000
```

4. API endpoints

Method	Path	Purpose
GET	/	Main UI
POST	/generate-comic	Create title and 5 panels (JSON)
POST	/generate-image	Create one panel image, returns image_url
POST	/download-pdf	Create PDF, returns pdf_url
GET	/comic-preview	Standalone preview page
GET	/export-success	Export success page

5. Using the app

- Enter story, character and setting; choose tone and art style.
- Click Generate Comic and wait for the 5 panels.
- Use Redraw on any panel you want to regenerate.
- Click Download PDF.

6. Known limitations

- Image generation depends on the Stability AI API and its credits.
- Redrawn panels may differ slightly in character look.
- Comics are fixed at 5 panels.

Communication

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Communication

Key messages and talking points for the demo.

Section	Key message	Approx. time
Opening	Anyone with a story can make a comic; no drawing skill needed	30 s
Problem	Artists are costly, drawing tools are hard, AI tools lose character consistency	45 s
Solution	Two-stage AI pipeline: Gemini writes the script, SDXL draws the panels	1 min
Live demo	Enter a story, generate, redraw one panel, export PDF	3 min
Tech	FastAPI, Pydantic, ReportLab; text kept out of images so bubbles stay readable	1 min
Close	Future scope and thanks	30 s

Communication tips

- Speak to the user problem first, technology second.
- Explain each screen as it appears.
- Have a backup recording in case an API call is slow.
- Share the speaking time evenly across team members.

Demonstration of Proposed Features

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Demonstration of Proposed Features

Step	Feature shown	What to do	Expected result
1	Input form	Enter a story, character, setting; choose tone and art style	Form accepts values
2	Script generation	Click Generate Comic	Progress bar, then title and 5 panels of text
3	AI artwork	Wait for images	Each panel shows its image
4	Dialogue bubbles	Look at the panels	Dialogue drawn over the image
5	Panel redraw	Click Redraw on one panel	Only that panel changes
6	PDF export	Click Download PDF	Storyboard PDF is created and opens
7	Different story	Run a second story with a different tone and style	App is not limited to one example

Sample input (edit as you like)

Story: A lighthouse keeper discovers that the light attracts friendly sea creatures.
 Character: Mira, a young keeper with a yellow raincoat
 Setting: A stormy island at night
 Tone: Heartwarming Art style: Watercolor

Project Demo Planning

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Project Demo Planning

Time	Activity	Owner
0:00	Introduction and problem	Arun N
1:00	Solution and architecture	Pushkaran P
2:30	Live demo	Pravin R
6:00	Code and tech walkthrough	Bhuvanesh P
8:00	Future scope, Q and A	Jayaraj / All

Pre-demo checklist

- Server starts with python -m uvicorn app.main:app.
- Both API keys are set in .env and have credit.
- Test story already run once; a backup screen recording is ready.
- Browser zoom and screen sharing checked.
- Sample PDF output saved in case of network trouble.

Scalability and Future Plan

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Scalability and Future Plan

Scalability

- Stateless FastAPI endpoints can run several Uvicorn workers or containers behind a load balancer.
- Image generation is the slowest step; panels can be generated in parallel or through a task queue.
- Move generated files from local static/ to cloud storage.
- Cache repeated prompts and add per-user rate limits.

Future plan

Phase	Feature
Next	Choose 4 to 8 panels; edit text before drawing; character reference image for better consistency
Later	User accounts and saved comic library; multiple pages; share link
Long term	Voice narration, animated panels, multiple languages, mobile app

Team Involvement in Demonstration

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Team Involvement in Demonstration

Roles below are a suggested split; adjust them to match who did what.

Member	Role in project	Role in demo
Arun N	Ideation, requirements, frontend	Introduction and problem
Pushkaran P	Gemini script generation, architecture	Solution and architecture
Pravin R	Image generation, routes	Live demo
Bhuvanesh P	PDF export, UI layout	Code and tech walkthrough
Jayaraj	Testing, documentation	Future scope, Q and A